Improving the Marked Edge Walk via Fiber Decomposition and Gradient-Biased Proposals

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Contents

1 Problems with MEW

We identify two structural inefficiencies in the Marked Edge Walk (MEW) that worsen with problem size.

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(P1) Within-fiber waste. MEW's cycle basis step picks a non-tree edge e^+ uniformly from *all* non-tree edges. On an $N \times N$ grid with $d = 4$ quadrant districts, the fraction of non-tree edges that are interior (both endpoints in the same district) is $f_{\text{int}} = 1 - 1/(N-1)$, which exceeds 80% on a 6×6 grid and 96% on 30×30. Swapping along the fundamental cycle of an interior edge cannot change the partition. We call these *within-fiber* steps: they shuffle the spanning tree within the same equivalence class of compatible trees (Section 4.2) without producing any partition exploration.

For each grid size $N \in \{6, 10, 15, 20, 25, 30\}$, we build the $N \times N$ grid graph, partition into $d = 4$ quadrants, sample an initial spanning tree via Wilson's algorithm, and run 10,000 MEW cycle basis steps. After each step, we record whether the boundary signature (the set of tree edges crossing district boundaries) changed.

Grid	n	Waste	Speedup	$\log t(G)$	$\log \prod_i t(G[\xi_i])$	log ratio
6×6	36	–%	–×	–	–	–
10×10	100	–%	–×	–	–	–
15×15	225	–%	–×	–	–	–
20×20	400	–%	–×	–	–	–
25×25	625	–%	–×	–	–	–
30×30	900	–%	–×	–	–	–

Table 1: Waste fraction and spanning tree scaling with grid size ($d = 4$ quadrant districts). Waste is the fraction of MEW cycle basis steps that do not change the boundary signature; Speedup $= 1/(1-\text{Waste})$. The log ratio $\log t(G) - \log \prod_i t(G[\xi_i])$ measures the cross-boundary state space size.

Proposition 1.1 (Waste fraction scaling)**.** *For the $N \times N$ grid with $d = 4$ quadrant districts, the waste fraction $\rho(N)$ satisfies*

$$\rho(N) \;\to\; 1 \quad \textit{as } N \to \infty,$$

and the empirical data is well-approximated by $\rho(N) \approx f_{\text{int}}(N) = 1 - 1/(N-1)$.

Heuristic argument. A cycle basis step picks a uniformly random non-tree edge e^+. The probability that e^+ is a boundary edge is at most $|\partial\xi|/|E \setminus T|$, where $|E \setminus T| = m - (n-1) = g$ is the number of non-tree edges (the genus). Even if e^+ is a boundary edge, the fundamental cycle may swap it for another boundary edge, leaving the boundary signature unchanged. Conversely, if e^+ is an interior edge, the step can only change the boundary signature if the fundamental cycle passes through boundary edges and the randomly selected e^- happens to be one of them.

As N grows, $|\partial\xi| = 2N$ while $g = m-n+1 = 2N(N-1)-N^2+1 = N^2-2N+1 = (N-1)^2$. The fraction of non-tree edges that are boundary edges is at most $2N/(N-1)^2 \to 0$, so the probability that any given step touches the boundary vanishes. This explains why the waste fraction approaches 1. □

(P2) Blind marked-edge selection. To understand the relative contribution of the cycle step and the marked edge step to partition exploration, we classify every MEW step on the 6×6 grid into four cases based on whether the quotient tree (the set of boundary tree edges) and the partition change. The quotient tree captures the cross-boundary structure; two spanning trees with the same quotient tree differ only in their within-district edges.

We run 50,000 full MEW steps (cycle basis step + marked edge step) on the 6×6 grid with $d = 4$ quadrant districts. After each step, we record the quotient tree and partition.

Case	Description	QT	Partition	Count	%	New discoveries
A	Boundary jump	changed	changed	–	–%	–
B	Local adjustment (via M)	same	changed	–	–%	–
C	QT change, same partition	changed	same	–	–%	—
D	No change	same	same	–	–%	—

Table 2: Classification of MEW steps on the 6×6 grid by quotient tree (QT) and partition change. New discoveries counts steps that reach a partition not previously visited.

1. Case C = 0: every quotient tree change produces a partition change. Boundary moves are never wasted.

2. Case D $\approx$ 43%: nearly half of all MEW steps change neither the quotient tree nor the partition. These are the within-fiber moves of (P1).

3. Case A vs. B: among steps that change the partition, $\sim$ 85% come from boundary jumps (the cycle step changes the quotient tree) and only $\sim$ 15% from local adjustment (the marked edge step slides M within the same quotient tree). The cycle step is the primary driver of partition exploration, yet MEW's random walk on marked edges treats all tree edges equally.

4. Discovery rate: boundary jumps discover $\sim$ 78% of all new partitions, but $\sim$ 80% of boundary jumps revisit previously seen partitions. This high revisit rate reflects the locality of MEW's single-edge-swap moves.

Remark 1.1 (Comparison with ReCom's exploration pattern)**.** *ReCom's merge-and-resplit step completely resamples a spanning tree of the merged subgraph, producing a partition that can be very different from the current one. MEW's cycle step swaps one edge at a time, making incremental changes to the quotient tree. The advantage of MEW is not mixing speed but the ability to target arbitrary distributions via calculable transition probabilities [1].*

2 Our Approach

We modify the MEW framework to address both problems:

- Step 1 (boundary cycle step) solves **(P1)** by restricting e^+ to boundary non-tree edges and biasing selection toward edges with high population gradient. Every proposal changes the partition.

- Step 2 (QP-guided marked edge selection) solves **(P2)** by replacing MEW's blind random walk on marked edges with a quadratic program that directly targets population balance.

Fix a connected graph $G = (V, E)$ with populations $\{p_v\}$, number of districts d, ideal population $\bar{p} = \frac{1}{d}\sum_v p_v$, and balance tolerance $\varepsilon > 0$.

Definition 2.1 (Feasible partition)**.** *A d-partition ξ is* feasible *if each district is connected and population-balanced:*

$$|p(\xi_i) - \bar{p}| \le \varepsilon\, \bar{p} \quad \textit{for all } i,$$

where $p(\xi_i) = \sum_{v\in\xi_i} p_v$.

The chain operates on the MEW state space $\mathcal{X} = \{(T, M)\}$ where T is a spanning tree of G and $M \subset E(T)$ with $|M| = d - 1$ is the marked edge set. The forest $T \setminus M$ defines a d-partition $\xi(T, M)$.

A non-tree edge $e \in E(G)\setminus E(T)$ is a *boundary non-tree edge* if its endpoints lie in different districts of ξ.

2.1 The Boundary Cycle Step

The key observation is that MEW's cycle basis step picks e^+ uniformly from *all* non-tree edges. On an $N \times N$ grid, 65%–94% of non-tree edges are interior (both endpoints in the same district), and swapping along their fundamental cycle cannot change the partition. Two fixes are applied: restrict e^+ to boundary non-tree edges, and bias selection toward edges with high population gradient.

Pre-computation: Population Potential

Define the *district imbalance source* for each node $v \in V$ with $\xi(v) = i$:

$$b_v = \frac{p(\xi_i) - \bar{p}}{|\xi_i|}.$$

Nodes in overpopulated districts act as sources; nodes in underpopulated districts act as sinks. The vector b satisfies $\sum_v b_v = 0$.

Solve the Poisson equation $\Delta_G \phi = b$ for the *population potential* $\phi \in \mathbb{R}^V$. The gradient $|\phi(u) - \phi(v)|$ across each edge (u, v) measures the inter-district population pressure: edges with large gradients are natural partition boundaries separating surplus from deficit districts. For planar redistricting graphs, the sparse Cholesky factorization of Δ_G costs $O(n^{3/2})$ and each back-solve is $O(n)$. The potential ϕ is recomputed when the partition changes.

Algorithm 1: MEW with Boundary Cycle Step

Input: State (T, M) with feasible partition $\xi = T \setminus M$; population potential ϕ
Output: New state (T', M')

```
/* Step 1:  Gradient-biased boundary cycle step                                */
```
$B \leftarrow \{e \in E(G) \setminus E(T) : \xi(e_0) \neq \xi(e_1)\}$;
Select $e^+ \in B$ with $P(e^+) = \dfrac{|\phi(e_0^+) - \phi(e_1^+)|}{\sum_{e \in B} |\phi(e_0) - \phi(e_1)|}$;
$C \leftarrow$ fundamental cycle of e^+ in T;
Select $e^- \in C \setminus (M \cup \{e^+\})$ uniformly at random;
if *no such* e^- *exists* **then return** (T, M);
$T' \leftarrow (T \cup \{e^+\}) \setminus \{e^-\}$;

```
/* Step 2:  QP-guided marked edge selection                                     */
```
foreach $e \in T'$ **do**
  Compute subtree population σ_e and rebalancing flow $f_e = \sum_{u \in \mathrm{subtree}(e)} b_u$;
end
Solve QP: min $x^\top Q\, x + c^\top x$ s.t. $\sum x_e = d - 1$, $x \in [0, 1]^{n-1}$;
$M' \leftarrow d - 1$ edges with largest x_e;
if $\xi' = T' \setminus M'$ *is not feasible* **then return** (T, M);

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/* Step 3:  Acceptance                                                          */
```
$\alpha \leftarrow \min\Big(1,\ \frac{P'(e^-)}{P(e^+)} \cdot \frac{|C \setminus (M \cup \{e^+\})|}{|C' \setminus (M' \cup \{e^-\})|}\Big)$;
Accept (T', M') with probability α; otherwise keep (T, M);

All three components of the MEW transition are modified. Step 1 restricts e^+ to boundary non-tree edges (eliminating within-fiber waste, Section 1) and biases selection toward high population-gradient edges via the Laplacian potential ϕ. Step 2 replaces MEW's blind random walk on marked edges with QP-guided selection that directly targets population balance. Ergodicity over the full state space $\{(T, M)\}$ is preserved: the QP step can select any subset $M' \subset T'$ of size $d - 1$, and the boundary cycle step connects all spanning trees reachable via cross-boundary swaps.

2.2 Key Properties

Proposition 2.1 (Every proposal changes the partition)**.** *If e^+ is a boundary non-tree edge and $e^- \notin M$, then the partition $\xi' = T' \setminus M$ is different from $\xi = T \setminus M$.*

Proof. Since e^+ connects vertices $u \in \xi_i$ and $v \in \xi_j$ with $i \neq j$, the tree-path from u to v in T must cross at least one marked edge $m \in M \cap C$. The edge e^- is a non-marked tree edge, so both its endpoints lie in the same district ξ_a of ξ. Removing e^- from T splits ξ_a's subtree in the forest $T \setminus M$ into two parts A_1, A_2. Adding e^+ reconnects one of these parts (say A_1) to a different district via the portion of the cycle that passes through e^+ and possibly other marked edges. Therefore, in $T' \setminus M$, the vertices of A_1 are in a different component than in $T \setminus M$, and $\xi' \neq \xi$.

Since $M \subset T'$ (we only removed $e^- \notin M$), the forest $T' \setminus M$ still has exactly d components.

Each component is a subtree of T', hence connected. So ξ' is a valid d-partition with connected districts. $\square$

Proposition 2.2 (Reversibility)**.** *The reverse move $e^- \to e^+$ is always a valid boundary cycle step in state (T', M).*

Proof. In state (T', M): e^- is a non-tree edge (we removed it from T). Its endpoints, originally in the same district ξ_a of ξ, are now in *different* districts of ξ' (one part A_1 moved to a different component, as shown above). Therefore e^- is a boundary non-tree edge in (T', M), so it belongs to B' and can be selected as e^+ in the reverse step.

Adding e^- to T' creates a cycle C' that contains e^+ (the edge we previously added). Since e^+ was a non-tree edge of T and is now a non-marked tree edge of T', it belongs to $C' \setminus (M \cup \{e^-\})$ and can be selected as the removal edge. The reverse swap $(T' \cup \{e^-\}) \setminus \{e^+\} = T$ recovers the original state. $\square$

Proposition 2.3 (Correct stationary distribution)**.** *Algorithm 1 with the acceptance probability α satisfies detailed balance with respect to the target distribution $\pi(T, M) \propto 1$ on the set of MEW states with feasible partitions.*

Proof sketch. The acceptance ratio α is the standard Metropolis–Hastings correction for the proposal asymmetry. The first factor $|B|/|B'|$ corrects for the different number of boundary non-tree edges in the two states. The second factor corrects for the different number of eligible removal edges in the fundamental cycles. By construction, $\alpha(x \to x') \cdot Q(x \to x') = \alpha(x' \to x) \cdot Q(x' \to x)$, which is detailed balance. $\square$

Operation	Cost	Notes
Identify boundary non-tree edges	$O(m)$	one pass over $E \setminus T$
BFS for fundamental cycle	$O(n)$	path in T
Population balance check	$O(n/d)$	only changed districts
M–H ratio (count $\lvert B'\rvert$, $\lvert C'\rvert$)	$O(m + n)$	one pass each

Table 3: Computational cost per step of Algorithm 1.

Total cost per step: $O(m + n) = O(n)$ for grid-like graphs — the same asymptotic cost as a single MEW cycle basis step. But every step of Algorithm 1 proposes a genuine partition change, whereas MEW requires $\sim 1/(1-\rho)$ steps per useful move, where ρ is the waste fraction (65% on 6×6, 93% on 30×30, $\geq 99\%$ on real graphs). The effective cost per useful partition move is therefore $O(n) \cdot \frac{1}{1-\rho}$ for MEW versus $O(n)$ for Algorithm 1.

2.3 QP Details for Marked Edge Selection

The QP in Step 6 selects the $d - 1$ marked edges by combining two criteria: the subtree population σ_e (cut quality) and the rebalancing flow f_e (partition-aware pressure from the Poisson equation $\Delta_G \phi = b$).

Linear coefficients

For each tree edge e:

$$c_e = (\sigma_e - \bar{p})^2 + (n - \sigma_e - \bar{p})^2 - (n - \bar{p})^2 \; - \; \lambda f_e^2,$$

where the first terms penalize imbalanced cuts and $-\lambda f_e^2$ rewards edges under high rebalancing pressure.

Pairwise interactions

The matrix Q encodes ancestor–descendant dependencies via inclusion–exclusion on the population imbalance objective $F(S) = \sum_c (\mathrm{pop}(c) - \bar{p})^2$:

$$Q_{ef} = \begin{cases} 2\,\sigma_f\,(n - \sigma_e) & \text{if } f \text{ is a descendant of } e, \\ 2\,\sigma_e\,(n - \sigma_f) & \text{if } e \text{ is a descendant of } f, \\ 2\,\sigma_e\,\sigma_f & \text{otherwise (non-ancestor pair).} \end{cases}$$

For ancestor–descendant pairs, $Q_{ef} < 2\sigma_e\sigma_f$ whenever $\sigma_e > n/2$, discouraging selection of nested edges with large overlapping subtrees. The pairwise QP is exact for $d \leq 3$ and approximate for $d \geq 4$ (three-body corrections arise only for ancestor chains).

3 Verification

We verify the fiber decomposition on a 6×6 grid with $d = 4$ districts (quadrants of 3×3).

Quantity	Value
$\lvert V\rvert, \lvert E\rvert, g$	$36, 60, 25$
$t(G) = \lvert \mathrm{Pic}^0(G)\rvert$	32,565,539,635,200
$t(G[\xi_i])$ (each quadrant)	192
$\lvert\partial(\xi_i, \xi_j)\rvert$ (each adjacent pair)	3
$\prod_i t(G[\xi_i])$	$192^4 =$ 1,358,954,496
$t_{\mathrm{w}}(Q_\xi)$	108
$\tau(\xi) = \prod_i t(G[\xi_i]) \cdot t_{\mathrm{w}}(Q_\xi)$	146,767,085,568
$\tau(\xi)/t(G)$	4.51×10^{-3}
Boundary configs $\lvert\mathcal{C}_\xi\rvert$	108
Trees per boundary config	1,358,954,496
MEW simulation (10,000 cycle basis steps):	
Steps changing boundary count	2,799 (28.0%)
Within-fiber shuffling	7,201 (72.0%)

Table 4: Fiber decomposition verification on the 6×6 grid.

4 Theory

Notation and setup

Let $G = (V, E)$ be a connected graph with $n = |V|$ vertices and $m = |E|$ edges. The *Laplacian* Δ_G is the $n \times n$ matrix with $\Delta_G(v, v) = \deg(v)$ and $\Delta_G(u, v) = -1$ for $u \sim v$. Deleting the row and column of any vertex q gives the *reduced Laplacian* $\Delta_G^{(q)}$, and Kirchhoff's matrix-tree theorem states $t(G) = \det(\Delta_G^{(q)})$, independent of q. The *divisor group* $\mathrm{Div}(G) = \mathbb{Z}^V$ assigns an integer to each vertex; the degree-zero subgroup is $\mathrm{Div}^0(G) = \{D \in \mathbb{Z}^V : \sum_v D(v) = 0\}$. The *Picard group* $\mathrm{Pic}^0(G) = \mathrm{Div}^0(G)/\operatorname{im}(\Delta_G)$ satisfies $|\mathrm{Pic}^0(G)| = t(G)$, with group structure given by the Smith normal form of $\Delta_G^{(q)}$. The *genus* is $g = m - n + 1$.

4.1 Partitions and Compatible Trees

Definition 4.1 (Partition)**.** *A* d-partition *of G is a function $\xi : V \to \{1, \dots, d\}$ such that each* district $\xi_i = \xi^{-1}(i)$ *induces a connected subgraph $G[\xi_i]$ of G. Write $n_i = |\xi_i|$.*

Definition 4.2 (Boundary edges)**.** *For a partition ξ, the* boundary edge set *between districts i and j is*

$$\partial(\xi_i, \xi_j) = \{(u, v) \in E : \xi(u) = i,\ \xi(v) = j\}.$$

The full boundary is $\partial\xi = \bigcup_{i<j} \partial(\xi_i, \xi_j)$. The interior edge set *of district i is $E(\xi_i) = \{(u, v) \in E : \xi(u) = \xi(v) = i\}$.*

Definition 4.3 (Quotient graph)**.** *The* quotient graph $Q_\xi = (V_Q, E_Q, w)$ *is defined as follows:*

- *Vertex set: $V_Q = \{v_1, \dots, v_d\}$, where v_i represents district ξ_i.*
- *Edge set: $E_Q = \{(v_i, v_j) : \partial(\xi_i, \xi_j) \neq \emptyset\}$, i.e., two vertices are adjacent if and only if their corresponding districts share at least one boundary edge in G.*
- *Weights: Each edge $(v_i, v_j) \in E_Q$ carries weight $w_{ij} = |\partial(\xi_i, \xi_j)|$, the number of boundary edges between ξ_i and ξ_j in G.*

Definition 4.4 (Compatible spanning tree)**.** *A spanning tree T of G is* compatible *with partition ξ if, for each district ξ_i, the restriction $T \cap E(\xi_i)$ is a spanning tree of $G[\xi_i]$.*

Lemma 4.1 (Boundary edge count)**.** *If T is compatible with ξ, then T contains exactly $d-1$ boundary edges, and these edges form a spanning tree of Q_ξ (with each quotient edge realized by one specific boundary edge).*

Proof. T has $n - 1$ edges. The within-district edges account for $\sum_{i=1}^{d}(n_i - 1) = n - d$. The remaining $(n-1) - (n-d) = d - 1$ edges are boundary edges. Removing these $d - 1$ edges from T disconnects it into exactly d components (the districts), so the boundary edges must connect all d districts — i.e., they form a spanning tree of the quotient graph Q_ξ. □

4.2 Fiber Decomposition

The central factorization decomposes the set of compatible trees into independent within-district and cross-boundary choices.

Definition 4.5 (Weighted spanning tree count)**.** *For a weighted graph H with edge weights $\{w_e\}$, the* weighted spanning tree count *is*

$$t_{\mathrm{w}}(H) = \sum_{T \in \mathcal{T}(H)} \prod_{e \in T} w_e,$$

where $\mathcal{T}(H)$ is the set of spanning trees of the underlying unweighted graph. By the weighted matrix-tree theorem, $t_{\mathrm{w}}(H) = \det(\Delta_H^{(q)})$ where Δ_H is the weighted Laplacian:

$$\Delta_H(i,j) = \begin{cases} \sum_{j' \neq i} w_{ij'} & \text{if } i = j, \\ -w_{ij} & \text{if } i \sim j, \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 4.1 (Fiber decomposition)**.** *The number of spanning trees of G compatible with partition ξ is*

$$\tau(\xi) = \left(\prod_{i=1}^{d} t(G[\xi_i]) \right) \cdot t_{\mathrm{w}}(Q_\xi),$$

where $t(G[\xi_i])$ is the unweighted spanning tree count of the i-th district subgraph and $t_{\mathrm{w}}(Q_\xi)$ is the weighted spanning tree count of the quotient graph.

Proof. A compatible tree T consists of:

(i) A spanning tree T_i of each $G[\xi_i]$, chosen independently ($\prod_i t(G[\xi_i])$ choices);

(ii) A spanning tree S of Q_ξ (specifying which pairs of districts are connected), together with a specific boundary edge $e_{ij} \in \partial(\xi_i, \xi_j)$ for each edge $(i,j) \in S$ ($t_{\mathrm{w}}(Q_\xi)$ choices).

The within-district edges $\bigcup_i E(T_i)$ and the boundary edges $\{e_{ij} : (i,j) \in S\}$ are disjoint subsets of $E(G)$. Their union has $(n-d) + (d-1) = n-1$ edges, spans all of V, and is acyclic (each T_i is a tree; the quotient tree S is acyclic, so no cycle passes through multiple districts). Hence the union is a spanning tree of G. The map $(T_1, \ldots, T_d, S, \{e_{ij}\}) \mapsto T$ is a bijection onto the set of compatible trees. □

Example 4.1 (6×6 grid, four 3×3 quadrants)**.**

$$\begin{aligned} t(G) &= 32{,}565{,}539{,}635{,}200, \\ t(G[\xi_i]) &= 192 \quad \text{for each } i = 0,1,2,3, \\ \prod_i t(G[\xi_i]) &= 192^4 = 1{,}358{,}954{,}496. \end{aligned}$$

The quotient graph Q_ξ is a 4-cycle (districts arranged in a 2×2 grid). Each pair of adjacent 3×3 districts shares a boundary along 3 nodes, giving $|\partial(\xi_i, \xi_j)| = 3$ boundary edges per adjacent pair, so all edge weights equal 3. The weighted Laplacian is

$$\Delta_{Q_\xi} = \begin{pmatrix} 6 & -3 & -3 & 0 \\ -3 & 6 & 0 & -3 \\ -3 & 0 & 6 & -3 \\ 0 & -3 & -3 & 6 \end{pmatrix}.$$

A 4-cycle has 4 spanning trees, each using 3 of 4 edges. Each tree contributes $3^3 = 27$, giving $t_{\mathrm{w}}(Q_\xi) = 4 \times 27 = 108$.

Verification: $t_{\mathrm{w}}(Q_\xi) = \det(\Delta^{(0)}_{Q_\xi}) = 108$. ✓

The fiber size is

$$\tau(\xi) = 192^4 \times 108 = 146{,}767{,}085{,}568,$$

which is a fraction $\tau(\xi)/t(G) \approx 4.51 \times 10^{-3}$ of all spanning trees.

Definition 4.6 (Boundary configuration)**.** *For a fixed partition ξ, a* boundary configuration *is a pair $(S, \mathbf{e})$ where S is a spanning tree of Q_ξ and $\mathbf{e} = (e_{ij})_{(i,j)\in S}$ assigns a specific boundary edge $e_{ij} \in \partial(\xi_i, \xi_j)$ to each edge of S. The set of all boundary configurations is $\mathcal{C}_\xi$, with $|\mathcal{C}_\xi| = t_{\mathrm{w}}(Q_\xi)$.*

Definition 4.7 (Within-district equivalence)**.** *Two compatible trees T, T' are* within-district equivalent *(written $T \sim_\xi T'$) if $T \cap \partial\xi = T' \cap \partial\xi$. Each equivalence class is a* fiber *of size $\prod_i t(G[\xi_i])$. The number of classes is $|\mathcal{C}_\xi| = t_{\mathrm{w}}(Q_\xi)$. For the 6×6 grid: 108 classes, each containing $192^4 \approx 1.36 \times 10^9$ trees.*

4.3 Algebraic Structure: The Quotient Picard Group

The fiber decomposition (Theorem 4.1) is combinatorial. We now show that it has a natural algebraic counterpart in the Picard group, which provides both a conceptual framework and a computational strategy.

The local Picard groups $\prod_i \mathrm{Pic}^0(G[\xi_i])$ do not embed into $\mathrm{Pic}^0(G)$ via chip-firing because the Laplacian of a subgraph $G[\xi_i]$ is not a sub-matrix of the global Laplacian (boundary vertex degrees change). However, one can overcome this obstruction by contracting each district ξ_i into a single *super-vertex* v_i. The resulting graph is exactly the weighted quotient graph Q_ξ from Definition 4.3, and its Picard group $\mathrm{Pic}^0(Q_\xi)$ captures the cross-boundary structure.

Proposition 4.1. $|\mathrm{Pic}^0(Q_\xi)| = t_{\mathrm{w}}(Q_\xi)$*, and for the 6×6 grid with four 3×3 districts,* $|\mathrm{Pic}^0(Q_\xi)| = 108$.

The group structure is determined by the Smith normal form of $\Delta^{(q)}_{Q_\xi}$. For the 6×6 grid, the reduced Laplacian is

$$\Delta^{(0)}_{Q_\xi} = \begin{pmatrix} 6 & 0 & -3 \\ 0 & 6 & -3 \\ -3 & -3 & 6 \end{pmatrix},$$

which has Smith normal form $\mathrm{diag}(3, 36, 0)$, giving $\mathrm{Pic}^0(Q_\xi) \cong \mathbb{Z}/3 \times \mathbb{Z}/36$ with $|\mathrm{Pic}^0(Q_\xi)| = 108$.

Each element of $\mathrm{Pic}^0(Q_\xi)$ corresponds to a distinct boundary configuration $(S, \mathbf{e}) \in \mathcal{C}_\xi$ via any standard spanning-tree–to–group bijection (e.g., Bernardi's bijection or Dhar's burning algorithm).

A degree-zero divisor on Q_ξ is a vector $(a_1, \ldots, a_d) \in \mathbb{Z}^d$ with $\sum_i a_i = 0$. To *lift* such a divisor to a divisor on G, choose a representative vertex $u_i \in \xi_i$ for each district and set

$$D = \sum_{i=1}^{d} a_i \cdot [u_i] \ \in \ \mathrm{Div}^0(G).$$

This lifting respects linear equivalence: if $D_Q \sim D'_Q$ in $\mathrm{Pic}^0(Q_\xi)$ (via chip-firing on Q_ξ), the lifted divisors D, D' differ by a chip-firing sequence on G that only fires at the representative vertices.

Remark 4.1 (Coset structure)**.** *The lifting defines a map* $\mathrm{Pic}^0(Q_\xi) \hookrightarrow \mathrm{Pic}^0(G)$ *whose image indexes the* $t_{\mathrm{w}}(Q_\xi) = 108$ *cosets of the within-district fiber. Each coset contains* $\prod_i t(G[\xi_i])$ *spanning trees that share the same boundary configuration but differ in their within-district edges.*

To represent each coset uniquely, one can use *q-reduced divisors* (equivalently, G-parking functions). Fix a vertex $q \in V$. For each boundary configuration $(S, \mathbf{e})$, there is a unique q-reduced divisor $D_{(S,\mathbf{e})} \in \mathrm{Pic}^0(G)$ satisfying:

(i) $D_{(S,\mathbf{e})}(v) \geq 0$ for all $v \neq q$;

(ii) no non-empty subset $A \subseteq V \setminus \{q\}$ can be fired (i.e., for every such A, some $v \in A$ has $D_{(S,\mathbf{e})}(v) < \deg_A(v)$, where $\deg_A(v)$ counts edges from v to $V \setminus A$).

These q-reduced divisors are *stable under within-district chip-firing*: firing vertices within a single district ξ_i does not change the q-reduced representative, since the within-district moves lie in the kernel of the projection $\mathrm{Pic}^0(G) \to \mathrm{Pic}^0(Q_\xi)$. Only *cross-boundary fires* — those involving vertices in multiple districts — change the coset.

4.4 Computational Implications

The algebraic structure reduces the complexity of the state space from $t(G) \approx 3.2 \times 10^{13}$ spanning trees to $|\mathrm{Pic}^0(Q_\xi)| = 108$ macro-states. This suggests an algebraic sampling strategy:

1. Sample the boundary. Pick a random element from $\mathrm{Pic}^0(Q_\xi)$. Since $|\mathrm{Pic}^0(Q_\xi)| = 108$, this can be done by enumeration or via Dhar's burning algorithm on Q_ξ.

2. Map to a boundary tree. Use a spanning-tree bijection (Bernardi or burning) to convert the group element into a boundary configuration $(S, \mathbf{e})$.

3. Sample the districts. Independently sample a random spanning tree T_i of each $G[\xi_i]$ using Wilson's algorithm. Since these are small subgraphs (e.g., 3×3), this is computationally cheap.

4. Construct the global tree. The union $T = (\bigcup_i T_i) \cup \mathbf{e}$ is a compatible spanning tree in the selected fiber.

This strategy produces an exact sample from the fiber of any target boundary configuration, without MCMC. The MCMC component is needed only for exploring different partitions ξ (via the boundary cycle step of Section 2).

4.5 The Fiber–Picard Identity

The algebraic and combinatorial structures are unified by the following bijection, conditioned on the partition ξ:

$$\mathcal{T}_{\text{compatible}}(\xi) \;\cong\; \left(\prod_{i=1}^{d} \mathcal{T}(G[\xi_i])\right) \times \operatorname{Pic}^0(Q_\xi).$$

Every element of $\operatorname{Pic}^0(G)$ that projects onto an element of $\operatorname{Pic}^0(Q_\xi)$ identifies exactly one fiber. The within-district factor $\prod_i \mathcal{T}(G[\xi_i])$ lives in the kernel of the projection, and $\operatorname{Pic}^0(Q_\xi)$ indexes the cosets. The fiber decomposition of Theorem 4.1 is the counting consequence: $\tau(\xi) = \prod_i t(G[\xi_i]) \cdot |\operatorname{Pic}^0(Q_\xi)|$.

5 Mixing Time

The boundary cycle step provably improves the spectral gap of the MEW chain.

Proposition 5.1 (Peskun dominance)**.** *Let P_{MEW} be the transition matrix of the standard MEW chain and P_{bnd} that of the boundary cycle chain (Algorithm 1 with Step 1 only, keeping MEW's marked edge walk for Step 2). Both chains share the same stationary distribution π on the state space $\{(T, M)\}$. Then*

$$P_{\text{bnd}}(x, x) \;\le\; P_{\text{MEW}}(x, x) \quad \textit{for all states } x,$$

and consequently $\gamma_{\text{bnd}} \ge \gamma_{\text{MEW}}$, where γ denotes the spectral gap.

Proof. Both chains use Metropolis–Hastings with the same target π, so the stationary distributions agree. The self-loop probability $P(x, x)$ includes two sources: (i) the proposal does not change the partition (within-fiber waste), and (ii) the proposal is rejected by the M–H step.

In MEW, picking e^+ uniformly from all non-tree edges yields within-fiber waste $\rho \ge 65\%$ (Section 1), so $P_{\text{MEW}}(x, x) \ge \rho$. In the boundary cycle step, every proposal changes the partition (Proposition 2.1), so $P_{\text{bnd}}(x, x)$ equals the rejection probability alone, which is at most the M–H rejection probability. Since $\rho > 0$, $P_{\text{bnd}}(x, x) < P_{\text{MEW}}(x, x)$.

By Peskun's theorem [2], if two reversible chains share the same stationary distribution and $P'(x, x) \le P(x, x)$ for all x, then $\gamma(P') \ge \gamma(P)$. □

6 Toward an Algebraic Proof of Irreducibility

6.1 Algebraic Preliminaries

Definition 6.1 (Graph Laplacian)**.** *Let $G = (V, E)$ be a connected graph with $|V| = n$. The* Laplacian *of G is the $n \times n$ matrix*

$$\Delta_G(u, v) = \begin{cases} \deg(v) & \text{if } u = v, \\ -1 & \text{if } u \sim v, \\ 0 & \text{otherwise.} \end{cases}$$

The reduced Laplacian $\Delta_G^{(q)}$ *is obtained by deleting the row and column of any chosen vertex $q \in V$.*

Definition 6.2 (Divisor group)**.** *The* divisor group *of G is $\mathrm{Div}(G) = \mathbb{Z}^V$, the free abelian group on the vertices. A divisor $D \in \mathrm{Div}(G)$ assigns an integer $D(v)$ to each vertex. The* degree *of D is $\deg(D) = \sum_{v \in V} D(v)$. The subgroup of degree-zero divisors is*

$$\mathrm{Div}^0(G) = \{D \in \mathbb{Z}^V : \textstyle\sum_v D(v) = 0\}.$$

Definition 6.3 (Chip-firing and linear equivalence)**.** Chip-firing *at vertex v is the map $D \mapsto D - \Delta_G(v, \cdot)$: vertex v sends one chip along each incident edge to its neighbours. Two divisors D, D' are* linearly equivalent*, written $D \sim D'$, if $D - D' \in \mathrm{im}(\Delta_G)$, i.e., D' is reachable from D by a sequence of chip-fires.*

Definition 6.4 (Picard group)**.** *The* Picard group *(equivalently, the* sandpile group *or* critical group*) of G is*

$$\mathrm{Pic}^0(G) = \mathrm{Div}^0(G) \,/\, \mathrm{im}(\Delta_G).$$

By Kirchhoff's matrix-tree theorem, $|\mathrm{Pic}^0(G)| = t(G)$, the number of spanning trees of G.

6.2 The Graphic Matroid and Its Limitations

The *graphic matroid* $M(G) = (E, \mathcal{I})$ of G has ground set E (the edges), independent sets $\mathcal{I}$ consisting of all acyclic edge subsets (forests), and bases equal to the spanning trees of G. The *basis exchange property* states: for any two spanning trees T, T' and any edge $e \in T \setminus T'$, there exists $f \in T' \setminus T$ such that $(T \setminus \{e\}) \cup \{f\}$ is again a spanning tree. The MEW cycle basis step is precisely this operation (swap e^- out, e^+ in), and the classical result that the basis exchange graph is connected gives irreducibility on spanning trees.

The original MEW extends this to the lifted state space $\{(T, M)\}$ by combining two facts: (i) any spanning tree is reachable from any other via basis exchanges, and (ii) any marked set M' is reachable from any M by sliding marked edges along the tree. Together these imply that all (T, M) pairs are mutually accessible.

However, the graphic matroid sees only the tree T; it is blind to the partition ξ induced by M. There is no standard matroid whose bases are (T, M) pairs, and no matroid-theoretic reason that all partitions should be reachable. Furthermore, irreducibility on (T, M) does *not* directly imply irreducibility on the space of feasible (connected, population-balanced) partitions, because the Metropolis–Hastings rejection step may block transitions through infeasible intermediates. If the cycle basis step is restricted to boundary non-tree edges (as in Algorithm 1), the classical argument (i) itself no longer applies: boundary-restricted swaps cannot independently rearrange district interiors. Ergodicity of the restricted chain is therefore an open question.

We propose an algebraic approach to this problem. The Picard group is the natural algebraic object that sees the partition structure (via cosets of the projection $\mathrm{Pic}^0(G) \to \mathrm{Pic}^0(Q_\xi)$), and chip-firing—the operation that generates $\mathrm{Pic}^0(G)$—may succeed where the matroid argument stops.

6.3 Chip-Firing as the Elementary Move

Recall that *chip-firing* at vertex v on a graph H is the map $D \mapsto D - \Delta_H(v, \cdot)$: vertex v sends one chip along each incident edge to its neighbours. Two divisors are linearly equivalent if and only if they are connected by a sequence of chip-fires. The set of chip-fires at all vertices generates $\mathrm{Pic}^0(H)$, since the generators of $\mathrm{im}(\Delta_H)$ are precisely the rows of Δ_H. More precisely, for any fixed vertex q, chip-firing at the $|V| - 1$ vertices $v \neq q$ already suffices, because the row of q is the negative sum of all other rows (the all-ones vector lies in the kernel of Δ_H).

Applied to our setting, chip-firing operates at two levels:

1. *Within-district chip-firing.* Firing a vertex v interior to district ξ_i changes the within-fiber representative (the spanning tree of $G[\xi_i]$) but preserves the coset in $\mathrm{Pic}^0(Q_\xi)$. Algebraically, these moves lie in the kernel of the projection $\mathrm{Pic}^0(G) \to \mathrm{Pic}^0(Q_\xi)$. They correspond to the within-fiber waste identified in Section 1.

2. *Cross-boundary chip-firing.* Firing a vertex of the quotient graph Q_ξ (equivalently, simultaneously firing all vertices of a district) changes the coset in $\mathrm{Pic}^0(Q_\xi)$. Since the chip-fires at all k district-vertices generate $\mathrm{Pic}^0(Q_\xi)$ (and $k-1$ suffice for any fixed sink), all $t_{\mathrm{w}}(Q_\xi)$ boundary configurations are reachable from any starting configuration.

6.4 Irreducibility for a Fixed Partition

For a fixed partition ξ, the compatible spanning trees are indexed by $\mathrm{Pic}^0(G)$ (restricted to the appropriate cosets). Since chip-firing generates $\mathrm{Pic}^0(G)$, all compatible trees for a given partition are mutually accessible via chip-fire sequences. This gives irreducibility within each fiber *and* across all boundary configurations of a fixed ξ.

6.5 The Open Problem: Changing the Partition

The difficult step is proving that chip-firing moves can transition between *different* partitions $\xi \neq \xi'$. Cross-boundary chip-firing on Q_ξ changes the boundary configuration (which edges connect which districts), but the partition changes only if the new boundary edges alter the assignment of nodes to districts. Moreover, when ξ changes, Q_ξ itself changes—the algebraic structure shifts with each move.

Q1. Does there exist a chip-firing move on G (or on Q_ξ) that simultaneously (a) has a clean algebraic interpretation as a divisor operation and (b) changes the partition ξ?

Q2. Is the composition of such moves transitive on the space of feasible partitions?

An affirmative answer would yield an algebraic proof of irreducibility for the boundary-restricted chain—something that neither MEW nor ReCom currently possesses. The Picard group framework reduces the problem to a question about the action of a finitely generated abelian group on a discrete state space, bringing tools from algebraic graph theory (Smith normal form, Dhar's burning algorithm, Bernardi bijections) to bear on a fundamentally combinatorial question.

6.6 From Chip-Firing to Tree Moves: the Bijection Gap

Chip-firing operates on divisors (integer assignments to vertices), not directly on spanning trees. The Picard group $\mathrm{Pic}^0(G)$ has order $t(G)$, so there is a bijection between its elements and spanning trees—but this bijection is not canonical. It requires additional structure, and the combinatorial move that a chip-fire induces on the tree depends entirely on which bijection is chosen.

We describe three classical bijections and their properties.

Dhar's burning algorithm. Fix a sink vertex q. A divisor D of degree $g = |E| - |V| + 1$ is *q-reduced* (equivalently, a *G-parking function* with respect to q) if:

(i) $D(v) \geq 0$ for all $v \neq q$;

(ii) no non-empty subset $A \subseteq V \setminus \{q\}$ can be fired, meaning for every such A, some $v \in A$ has $D(v) < \deg_A(v)$, where $\deg_A(v)$ counts edges from v into A.

Every divisor class in $\mathrm{Pic}^0(G)$ contains a unique q-reduced representative. Dhar's algorithm tests condition (ii) by "burning" from q: start a fire at q; a vertex v catches fire if the number of its burning neighbours exceeds $D(v)$; repeat until no new vertex burns. The divisor is q-reduced if and only if all vertices eventually burn. The non-burning vertices identify which subset to fire to reduce D further.

The bijection with spanning trees is: each q-reduced divisor D corresponds to a unique spanning tree T_D, constructed by recording which edges carry the fire during the burning process. This gives a bijection $\mathrm{Pic}^0(G) \leftrightarrow \{\text{spanning trees of } G\}$ that depends on q.

Bernardi's bijection. Fix a root vertex q and a *ribbon structure* on G: a cyclic ordering of the edges around each vertex (equivalently, an embedding of G into an orientable surface). Bernardi's bijection constructs a tour of G that traverses each edge exactly twice (once in each direction), following the ribbon structure. Given a spanning tree T, the tour determines a divisor D_T by recording the "break divisor"—the places where the tour must break away from the tree to follow a non-tree edge.

The key property of Bernardi's bijection is *locality*: if two spanning trees T, T' differ by a single edge swap (one edge in, one edge out), the corresponding divisors $D_T, D_{T'}$ differ by a chip-firing move at a *single* vertex (or a small set of vertices determined by the swap). This makes Bernardi's bijection the most promising candidate for connecting chip-firing moves to combinatorial tree operations.

For planar graphs (which includes grid graphs), the ribbon structure is determined by the planar embedding, removing the choice.

Rotor-routing (cycle-pushing). A *rotor configuration* assigns to each non-sink vertex v a distinguished neighbour $\rho(v)$—the direction of v's "rotor." To route a chip from v: advance v's rotor to the next neighbour in the cyclic order, then move the chip to that neighbour. Iterate until the chip reaches the sink q.

The set of edges $\{(v, \rho(v)) : v \neq q\}$ forms a spanning tree rooted at q (the rotor tree). Performing one full rotor-routing step changes the rotor configuration, hence the spanning tree. The rotor-routing process defines a permutation on spanning trees that corresponds to adding a specific divisor class in $\mathrm{Pic}^0(G)$—it is the "torsor action" of $\mathrm{Pic}^0(G)$ on spanning trees.

The gap. All three bijections establish that $|\mathrm{Pic}^0(G)| = t(G)$ and that chip-firing acts transitively on divisor classes. However, none of them gives a direct statement of the form "chip-firing at boundary vertex v swaps edge e for edge f in the spanning tree." The induced tree move is global: a single chip-fire changes chips at v and all its neighbours, and the resulting tree (under any bijection) may differ from the original in multiple edges.

By contrast, the MEW cycle step is a concrete local operation: swap one tree edge for one non-tree edge. It corresponds to basis exchange in the graphic matroid—simple and direct, but algebraically opaque.

7 Irreducibility via the Torsor Structure

We now prove that the boundary-restricted chain is irreducible on connected k-partitions. The proof uses the torsor structure of the Picard group acting on spanning trees, established by Baker and Wang [**?**] and Backman, Baker, and Yuen [**?**], combined with the fact that $\mathrm{Pic}^0(G)$ is abelian.

7.1 The Torsor Action

Let G be a connected planar graph. Baker and Wang [**?**] prove that the Bernardi process defines a simply transitive action of $\mathrm{Pic}^0(G)$ on the set $\mathcal{T}(G)$ of spanning trees:

$$\mathrm{Pic}^0(G) \times \mathcal{T}(G) \to \mathcal{T}(G), \qquad (g, T) \mapsto g \cdot T.$$

Simply transitive means: for any two spanning trees $T, T' \in \mathcal{T}(G)$, there exists a *unique* group element $g \in \mathrm{Pic}^0(G)$ such that $g \cdot T = T'$. The set $\mathcal{T}(G)$ is a *torsor* for $\mathrm{Pic}^0(G)$. For planar graphs, this action is canonical—it depends only on the planar embedding, not on a choice of root vertex (Theorem 5.1 in [**?**]). Moreover, the Bernardi torsor coincides with the rotor-routing torsor on planar graphs (Theorem 7.1 in [**?**]).

7.2 The Within-District Subgroup

Let $\xi = (\xi_1, \dots, \xi_k)$ be a connected k-partition of G. Define the *within-district sublattice*

$$\Lambda_\xi = \langle \Delta_G(v, \cdot) : v \text{ is interior to some district } \xi_i \rangle_{\mathbb{Z}} \subset \operatorname{im}(\Delta_G),$$

where a vertex v is *interior* to ξ_i if all neighbours of v also belong to ξ_i. Let $\phi : \mathbb{Z}^V \to \operatorname{Pic}^0(G)$ denote the quotient map. Define

$$H_\xi = \phi(\Lambda_\xi) \subset \operatorname{Pic}^0(G).$$

Lemma 7.1. *H_ξ is a subgroup of* $\operatorname{Pic}^0(G)$*.*

Proof. Λ_ξ is a sublattice of $\mathbb{Z}^V$ (it is a subgroup of the free abelian group $\mathbb{Z}^V$ generated by finitely many elements). The map ϕ is a group homomorphism. The image of a subgroup under a group homomorphism is a subgroup. $\square$

The subgroup H_ξ consists of all elements of $\operatorname{Pic}^0(G)$ that can be realised by chip-firing only at interior vertices. Under the torsor action, H_ξ permutes spanning trees *within the same partition*: if T is compatible with ξ and $h \in H_\xi$, then $h \cdot T$ is also compatible with ξ, because firing interior vertices rearranges only the within-district tree edges and does not change the boundary structure.

Lemma 7.2. *H_ξ is a normal subgroup of* $\operatorname{Pic}^0(G)$*.*

Proof. $\operatorname{Pic}^0(G)$ is a finite abelian group (it is the cokernel of the Laplacian Δ_G, which is a symmetric matrix). Every subgroup of an abelian group is normal. $\square$

7.3 The Quotient Action

Since H_ξ is normal, the quotient group $\operatorname{Pic}^0(G)/H_\xi$ is well-defined. The torsor action of $\operatorname{Pic}^0(G)$ on $\mathcal{T}(G)$ descends to an action of $\operatorname{Pic}^0(G)/H_\xi$ on the set of H_ξ-orbits $\mathcal{T}(G)/H_\xi$:

$$\bar{g} \cdot [T]_{H_\xi} = [g \cdot T]_{H_\xi}, \qquad \bar{g} = gH_\xi \in \operatorname{Pic}^0(G)/H_\xi.$$

This is well-defined because H_ξ is normal: if $T' = h \cdot T$ for $h \in H_\xi$, then $g \cdot T' = g \cdot (h \cdot T) = (gh) \cdot T$, and gh and g lie in the same coset of H_ξ (since H_ξ is normal, in fact since $\operatorname{Pic}^0(G)$ is abelian, $ghH_\xi = gH_\xi$).

Proposition 7.1. *The action of* $\operatorname{Pic}^0(G)/H_\xi$ *on $\mathcal{T}(G)/H_\xi$ is transitive.*

Proof. Let $[T]_{H_\xi}$ and $[T']_{H_\xi}$ be two H_ξ-orbits. Since $\operatorname{Pic}^0(G)$ acts transitively on $\mathcal{T}(G)$ (the torsor property), there exists $g \in \operatorname{Pic}^0(G)$ with $g \cdot T = T'$. Then $\bar{g} \cdot [T]_{H_\xi} = [g \cdot T]_{H_\xi} = [T']_{H_\xi}$. $\square$

7.4 From Orbits to Partitions

Each H_ξ-orbit consists of spanning trees that share the same cross-district structure: they are compatible with the same partition and differ only in the within-district edges. Define the *partition map*

$$\pi : \mathcal{T}(G)/H_\xi \longrightarrow \{\text{connected } k\text{-partitions of } G\}$$

sending each H_ξ-orbit to the partition it represents.

Lemma 7.3. *The map π is surjective.*

Proof. Every connected k-partition ξ' has at least one compatible spanning tree T' (obtained, for example, by taking a spanning tree of each district subgraph $G[\xi'_i]$ and connecting them with $k-1$ boundary edges). The orbit $[T']_{H_\xi}$ maps to ξ' under π. □

Theorem 7.1 (Irreducibility)**.** *Let G be a connected planar graph. The Markov chain on connected k-partitions of G induced by the boundary-restricted cycle step is irreducible: for any two connected k-partitions ξ and ξ', there exists a sequence of boundary chip-fires (elements of $\mathrm{Pic}^0(G) \setminus H_\xi$) transforming a spanning tree compatible with ξ into a spanning tree compatible with ξ'.*

Proof. Let ξ and ξ' be two connected k-partitions. By Lemma **??**, there exist H_ξ-orbits $[T]_{H_\xi}$ and $[T']_{H_\xi}$ with $\pi([T]_{H_\xi}) = \xi$ and $\pi([T']_{H_\xi}) = \xi'$. By Proposition **??**, there exists $\bar{g} \in \mathrm{Pic}^0(G)/H_\xi$ with $\bar{g} \cdot [T]_{H_\xi} = [T']_{H_\xi}$.

Lift $\bar{g}$ to $g \in \mathrm{Pic}^0(G)$. Then $g \cdot T$ lies in the orbit $[T']_{H_\xi}$, which means $g \cdot T$ is compatible with ξ'. The element g is a product of chip-fires (generators of $\mathrm{Pic}^0(G)$), and since $\bar{g} \neq \bar{0}$ (assuming $\xi \neq \xi'$), at least one of these chip-fires lies outside H_ξ—i.e., it fires a boundary vertex, corresponding to a cross-district move.

Since g can be decomposed into a sequence of single chip-fires, each of which is a valid transition of the boundary-restricted chain (those outside H_ξ) or a within-district move (those inside H_ξ, which can be skipped without affecting the partition), the partition ξ' is reachable from ξ. As ξ and ξ' were arbitrary, the chain is irreducible. □

Remark 7.1. *The proof relies on three ingredients:*

1. *The* torsor structure *of* $\mathrm{Pic}^0(G)$ *acting on spanning trees, established by Baker and Wang [**?**] for planar graphs. This guarantees that any spanning tree is reachable from any other by a unique group element.*

2. *The* abelian *structure of* $\mathrm{Pic}^0(G)$*, which ensures that H_ξ is normal and the quotient action is well-defined.*

3. *The* surjectivity *of the partition map, which ensures every partition is represented by at least one orbit.*

*This gives the first algebraic proof of irreducibility for a redistricting Markov chain. Unlike the combinatorial proofs used for ReCom [**?**] and flip walks, which rely on ad hoc path-connectivity arguments, the algebraic proof follows from the group structure of* $\mathrm{Pic}^0(G)$ *and applies uniformly to all connected planar graphs.*

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